

Backend Developer Take-Home Assessment

Project: People Hub API

Experience: 0–1 Year (6 Months Preferred)

Duration: 3 Hours

Technology: Node.js, Express.js, MySQL

Objective

Develop a REST API for a simple People Hub application where administrators can manage departments and employees. No frontend is required.

Database

Table 1: departments

```
CREATE TABLE departments (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  departmentName VARCHAR(100) NOT NULL,  
  createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Table 2: employees

```
CREATE TABLE employees (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  employeeCode VARCHAR(20) UNIQUE,  
  fullName VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE,  
  mobile VARCHAR(15),  
  departmentId INT,  
  designation VARCHAR(100),  
  salary DECIMAL(10,2),  
  status ENUM('Active','Inactive') DEFAULT 'Active',  
  createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (departmentId) REFERENCES departments(id)  
);
```

Required APIs

- POST /departments - Create Department (no duplicates).
- GET /departments - List Departments.
- PUT /departments/:id - Update Department.
- DELETE /departments/:id - Prevent deletion if employees exist.
- POST /employees - Add Employee with validations.

- GET /employees - Pagination, search, department filter, status filter.
- GET /employees/:id - Employee details with department name.
- PUT /employees/:id - Update Employee.
- DELETE /employees/:id - Delete Employee.
- PATCH /employees/:id/status - Activate/Deactivate Employee.
- GET /dashboard - Employee and department statistics.

Employee Validations

- Employee Code must be unique
- Email must be unique
- Full Name is mandatory
- Department must exist
- Mobile should contain only digits

Bonus

Implement GET /employees/export to export employee data as CSV.

Expected Folder Structure

```
people-hub/
■■■ app.js
■■■ package.json
■■■ .env
■■■ config/db.js
■■■ routes/
■■■ controllers/
■■■ models/
■■■ middlewares/
```

Submission

- Source Code
- SQL Script
- Postman Collection
- README with setup instructions and API documentation

Evaluation

Criteria	Marks
Database Design	15
CRUD APIs	30
SQL Queries	15
Pagination & Search	15
Validation	10
Error Handling	10
Folder Structure & Code Quality	5
Bonus	10